COSC2637/2633 Big Data Processing
Lab 1: Tutorial/Lab class (Windows OS Users)

Objectives

- Access your individual jumphost that has already been setup for each student
- Create, access and then terminate an AWS EMR cluster
- Responsible use of cloud resources

Introduction

In order to use AWS EMR, RMIT ITS has created a relatively inexpensive jumphost on AWS for each student enrolled in the course. From this jumphost you will be to create, access and manage an EMR cluster. Creating a cluster requires allocation of additional physical machines in the AWS datacentre (which takes minutes to happen, depending on the size of cluster). However, as the EMR clusters are more expensive machines (and cost by time that the cluster is left set up), it is important that clusters are not left running after you have finished using them (and saved any output). So, at the end of a lab class, it is **very important** you **terminate** your cluster.

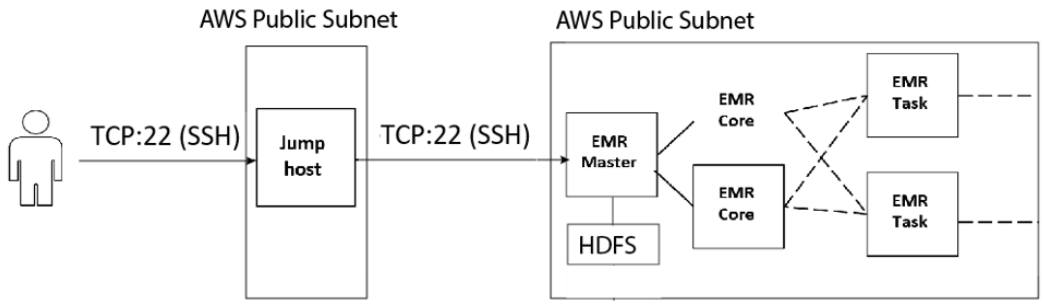
The jumphost and any EMR clusters launched will all be in the AWS US East Region (N.Virginia) (Also known as "us-east-1" or "Standard"). This means any clusters you create will have access to the Public Data Sets hosted in S3 that are provided by AWS in that region.

Example of student jumphost DNS: `sXXXXXXX.jump.cosc2637.route53.aws.rmit.edu.au`

Each student will have a personal SSH key and it will provide access to your jumphost and EMR cluster.

You can run `./create_cluster.sh` from your jumphost to launch your EMR cluster. After the script finishes running, there will be further instructions on how to access that specific cluster (Hadoop Master node).

Please **do not forget** to run `./terminate_cluster.sh` (from your jumphost) each time you have finished using your cluster (e.g., at the end of the lab class).



The steps to get you on to the AWS environment (Windows OS)

Keys (these should be emailed to you at or prior to your practical class)

1. Save the key location on your device,
2. Convert the key using PuTTYGen in Windows OS.
   (You may need to download and install PuTTY and PuTTYGen)

Converting Your Private Key Using PuTTYgen
PuTTY does not natively support the private key format (.pem) generated by Amazon EC2. PuTTY has a tool named PuTTYgen, which can convert keys to the required PuTTY format (.ppk). You must convert your private key into this format (.ppk) before attempting to connect to your jumphost account using PuTTY.

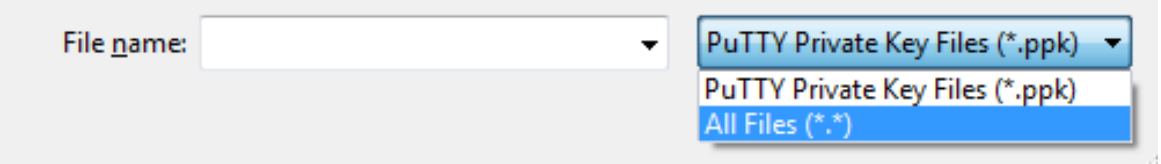
To convert your private key

a. Start PuTTYgen

b. Under **Type of key to generate**, choose **RSA**.



c. Choose **Load**. By default, PuTTYgen displays only files with the extension .ppk. To locate your .pem file, select the option to display files of all types.



d. Select your .pem, and then choose **Open**. Choose **OK** to dismiss the confirmation dialog box.

e. Choose **Save private key** to save the key in the format that PuTTY can use. PuTTYgen displays a warning about saving the key without a passphrase. Choose **Yes**.

   Note: A passphrase on a private key is an extra layer of protection, so even if your private key is discovered, it can't be used without the passphrase. The downside to using a passphrase is that it makes automation harder because human intervention is needed to log on to your jumphost account or copy files to your jumphost account.

f. Specify the same name for the key as the .pem file. PuTTYgen automatically adds the .ppk file extension.

Your private key is now in the correct format for use with PuTTY. You can now connect to your jumphost account ussing PuTTY's SSH client.

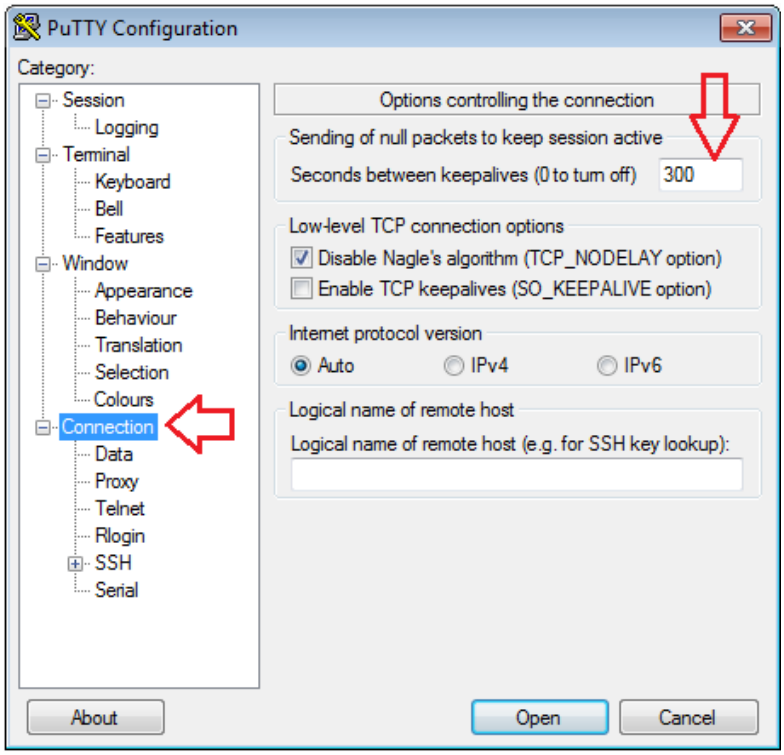
Links for more information:
http://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AccessingInstances.html

http://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AccessingInstancesLinux.html

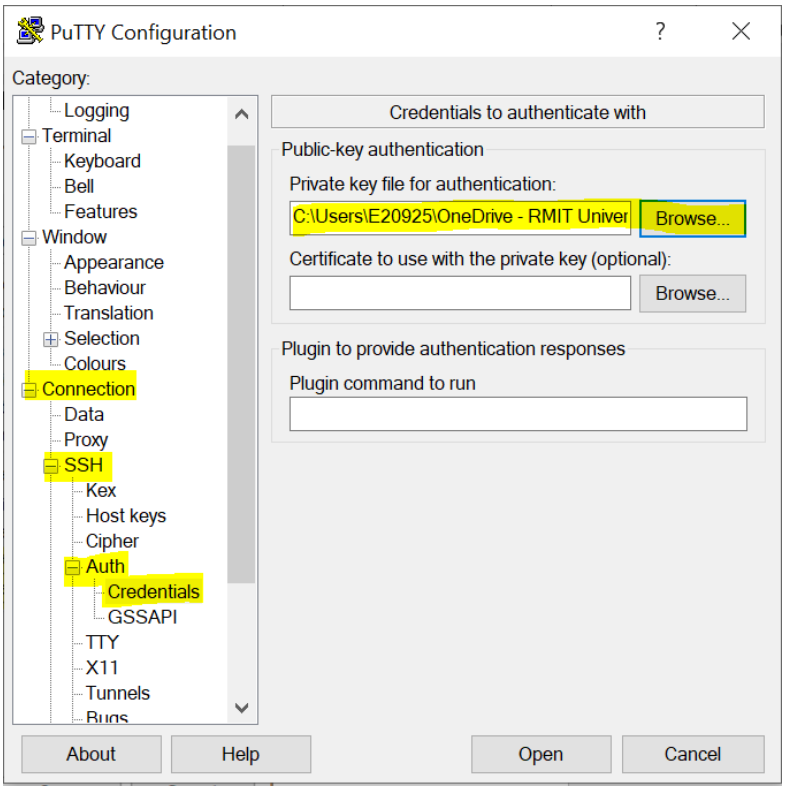
http://docs.aws.amazon.com/AWSEC2/latest/UserGuide/putty.html

3. Set up PuTTY connection

a. Configuration required to prevent the inactive shell phenomenon

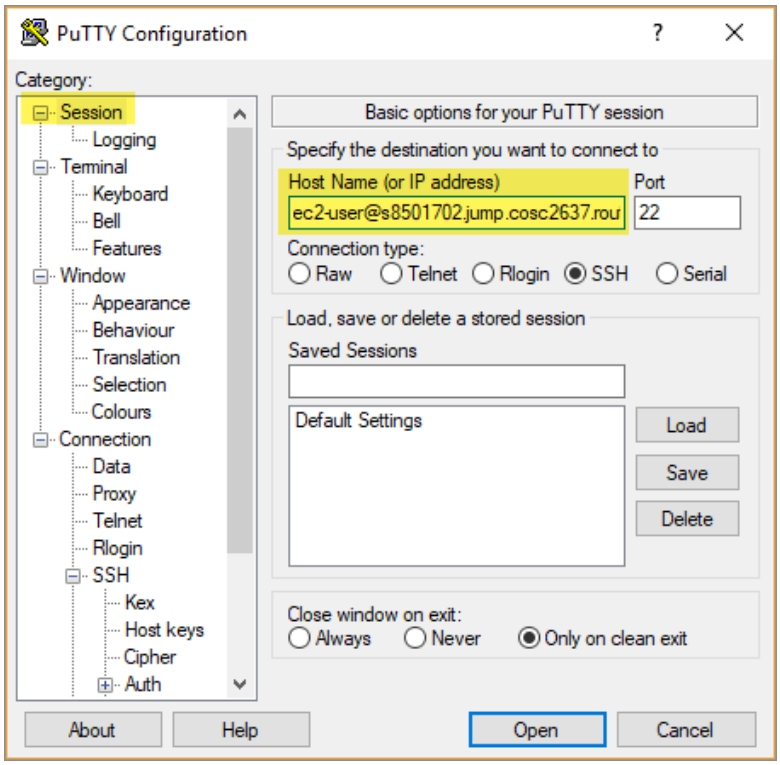


b. Add private key you saved earlier (extracted from the key pair file emailed to you)

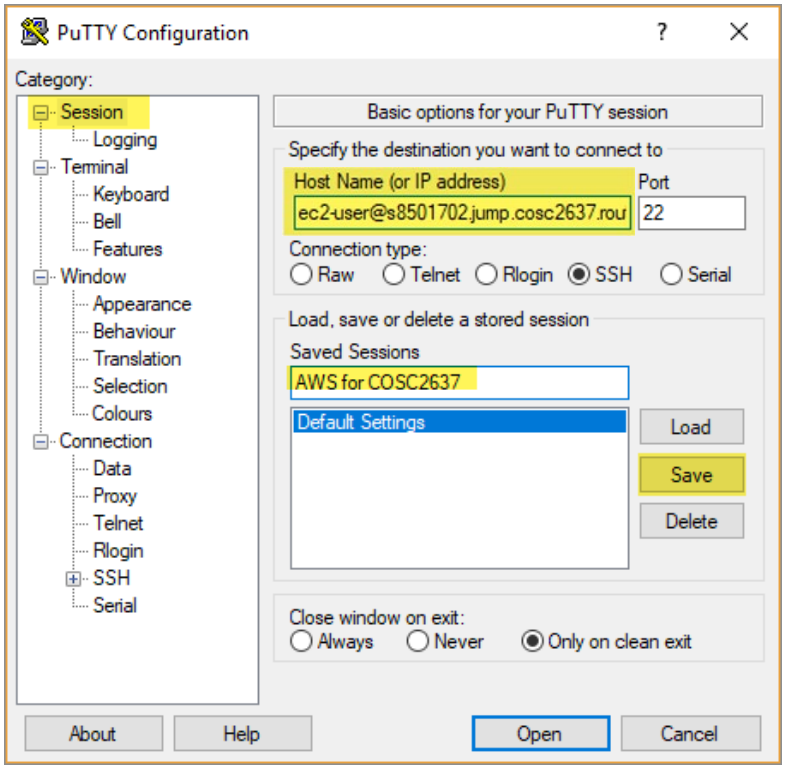


c. Enter jump host name:

`ec2-user@sXXXXXXX.jump.cosc2637.route53.aws.rmit.edu.au`
Where *sXXXXXXX* is your student number



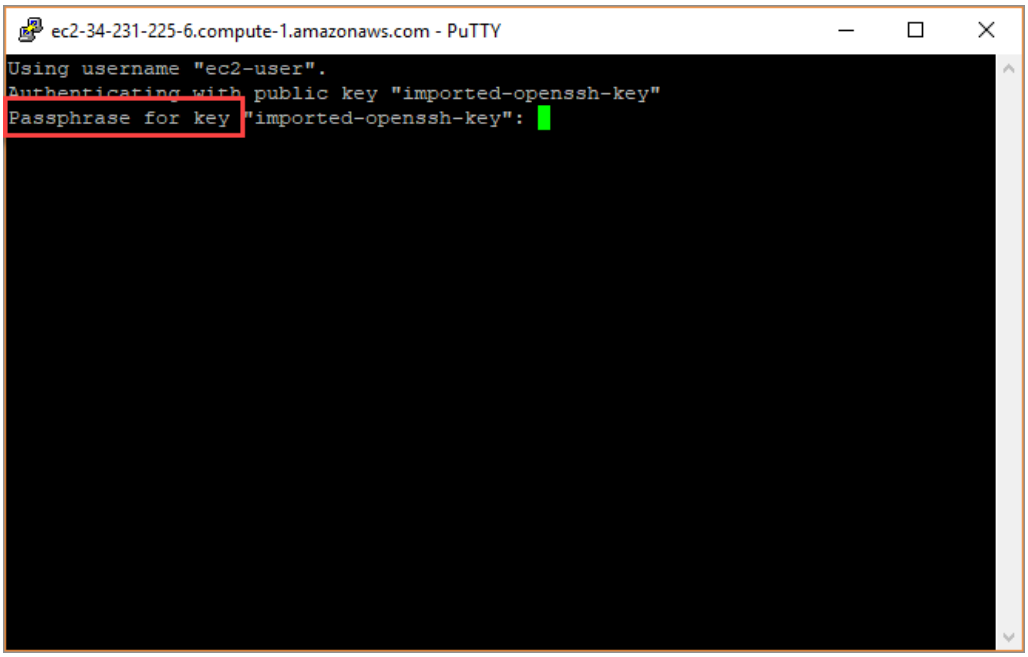
d. Save settings



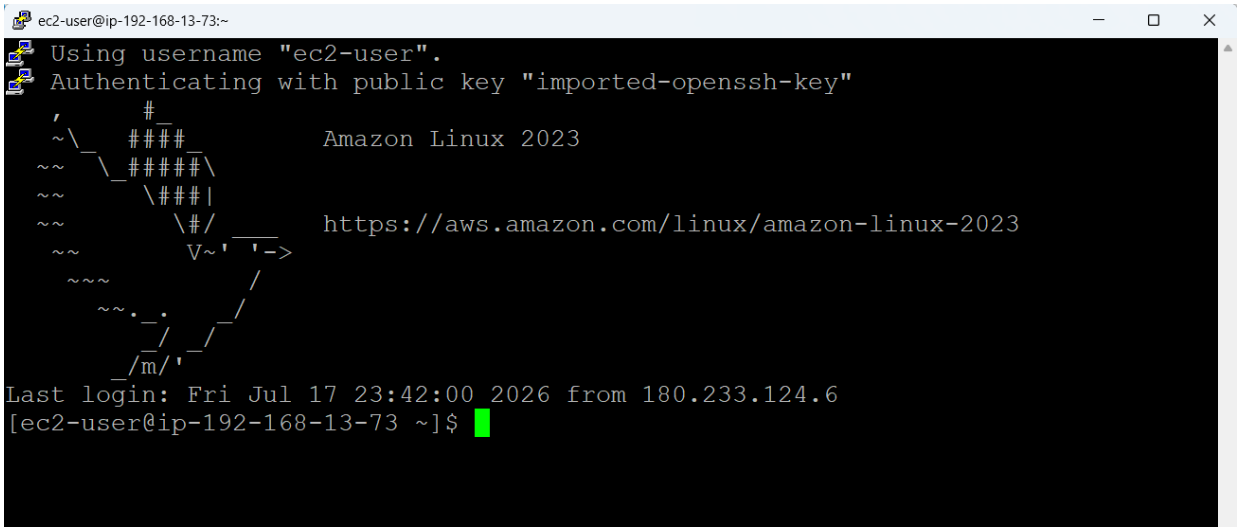
4. Open PuTTY connection

   Open button

If you entered a passphrase when saving the private key, you will be prompted for it now. This is why we advised “saving the key without a passphrase” (page 2, e).

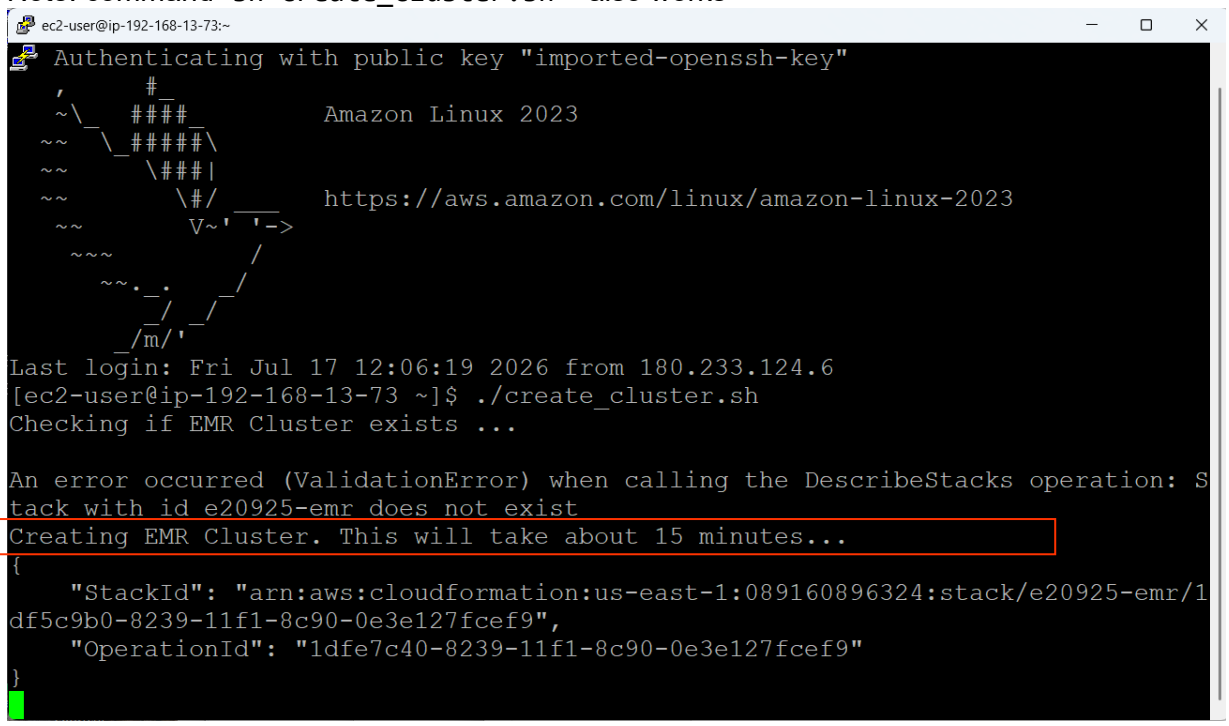


5. You should now have the AWS $ prompt



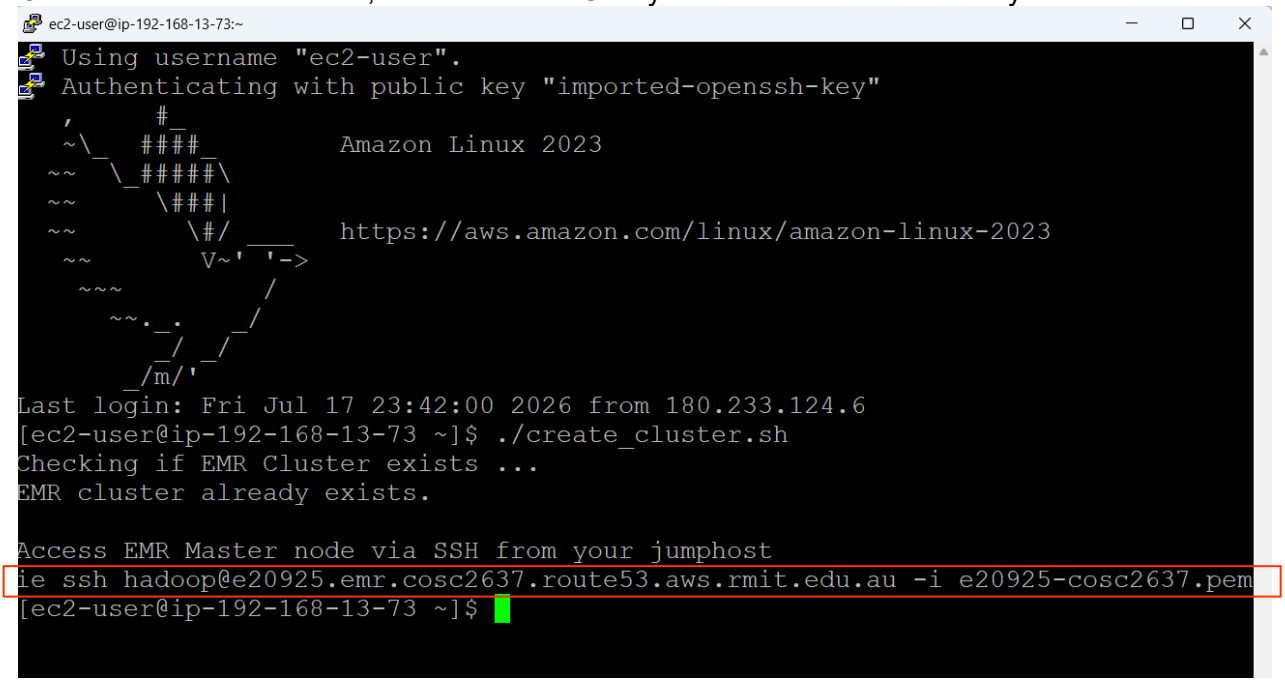
6. Enter `./create_cluster.sh`

   Note: command "`sh create_cluster.sh`" also works



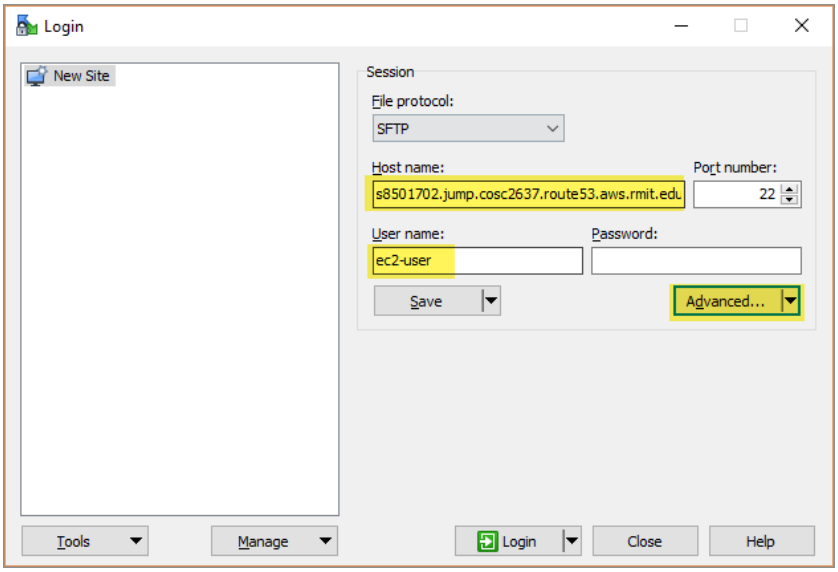
The EMR cluster will now be created. **It can take 15 minutes or more.** The script begins by checking whether a cluster already exists, so the ValidationError is expected as it is simply saying that the cluster doesn't already exist.

7. Once the cluster is created, it will show the URL you need use to connect to your cluster.

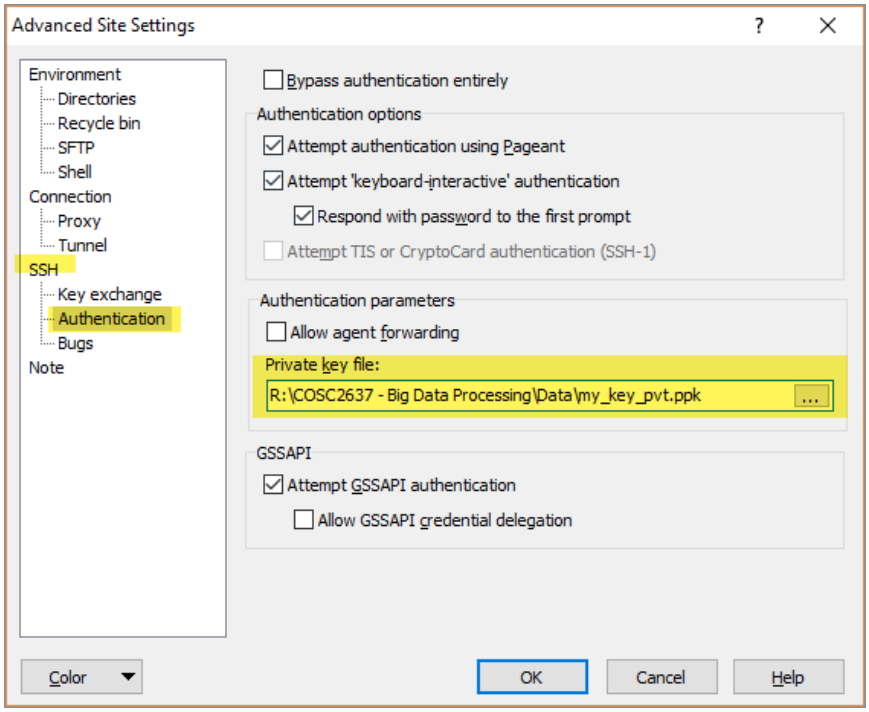


Now, using WinSCP (you may need download and install WinSCP) for transferring the key to JumpHost home (the key is xxx-xxx.pem file emailed to you at or prior to practical class)

1. Enter jump host name, user and Click Advanced, then select Advanced

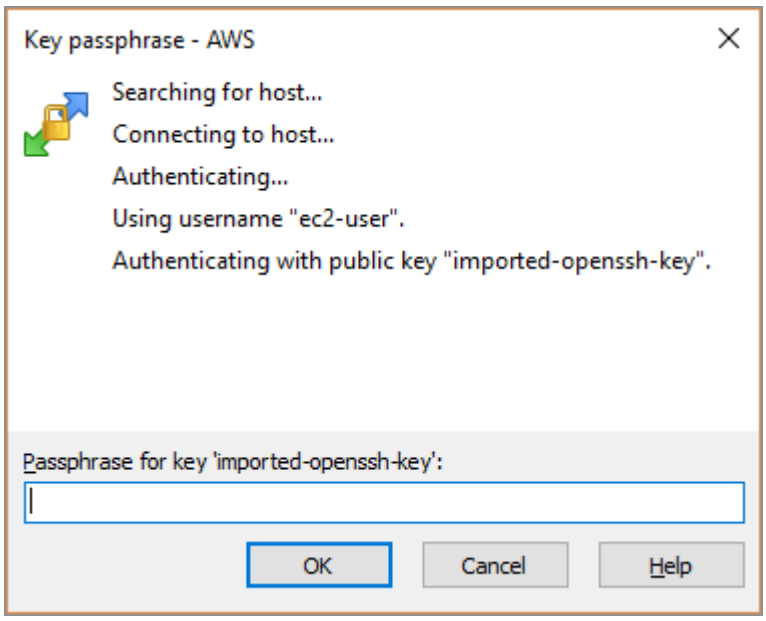


2. Under SSH |Authentication, browse to the private key you extracted from the emailed key pair

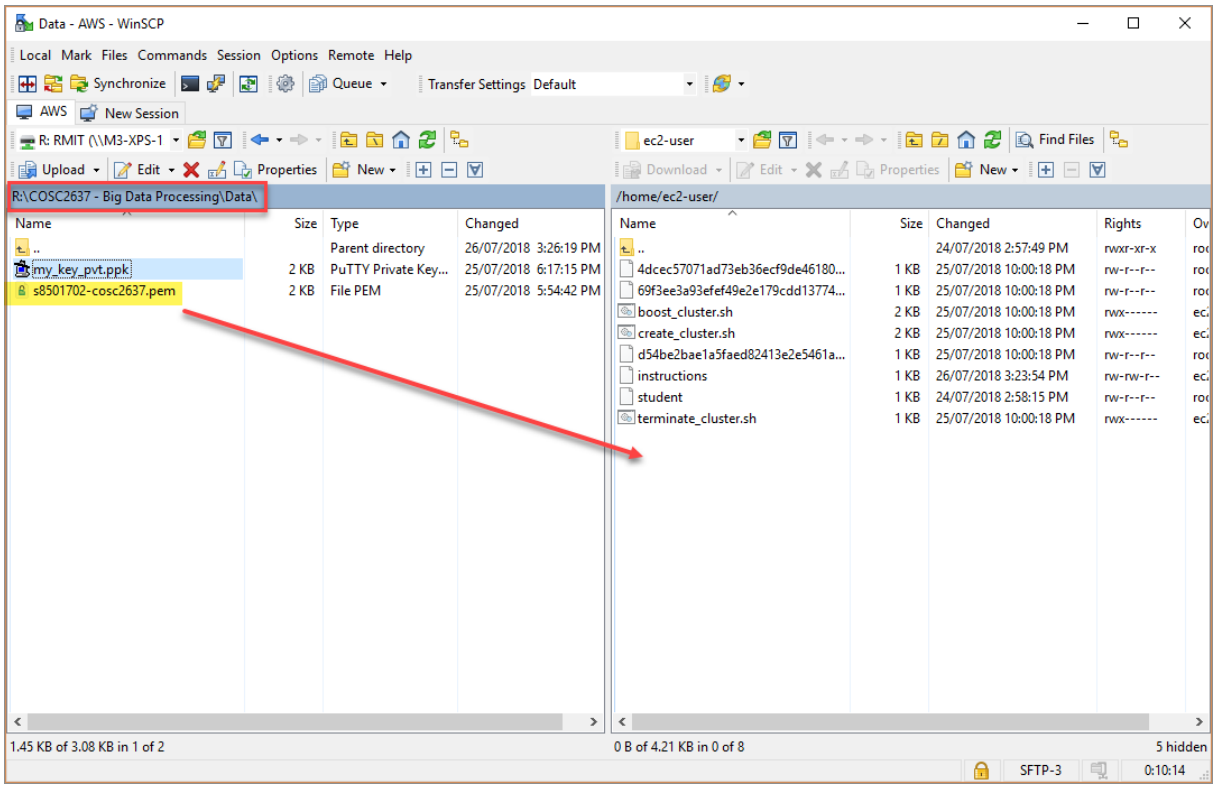


3. Click OK, and when you're back to the Login screen, save the connection if desired.

4. Login. If you entered a passphrase when saving the private key, you will be prompted for it now



5. In the left panel, browse to the location of the key pair file you were emailed (*.pem), then drag this to the right panel (the jump host), to copy the key pair file.



Then, go to JumpHost home (as shown in the last picture in page 5 of this document) and you should find xxx-xxx.pem. Next

$ chmod 400 xxx-xxx.pem

You will see -r-------- for xxx-xxx.pem file. It means file owner can read only.

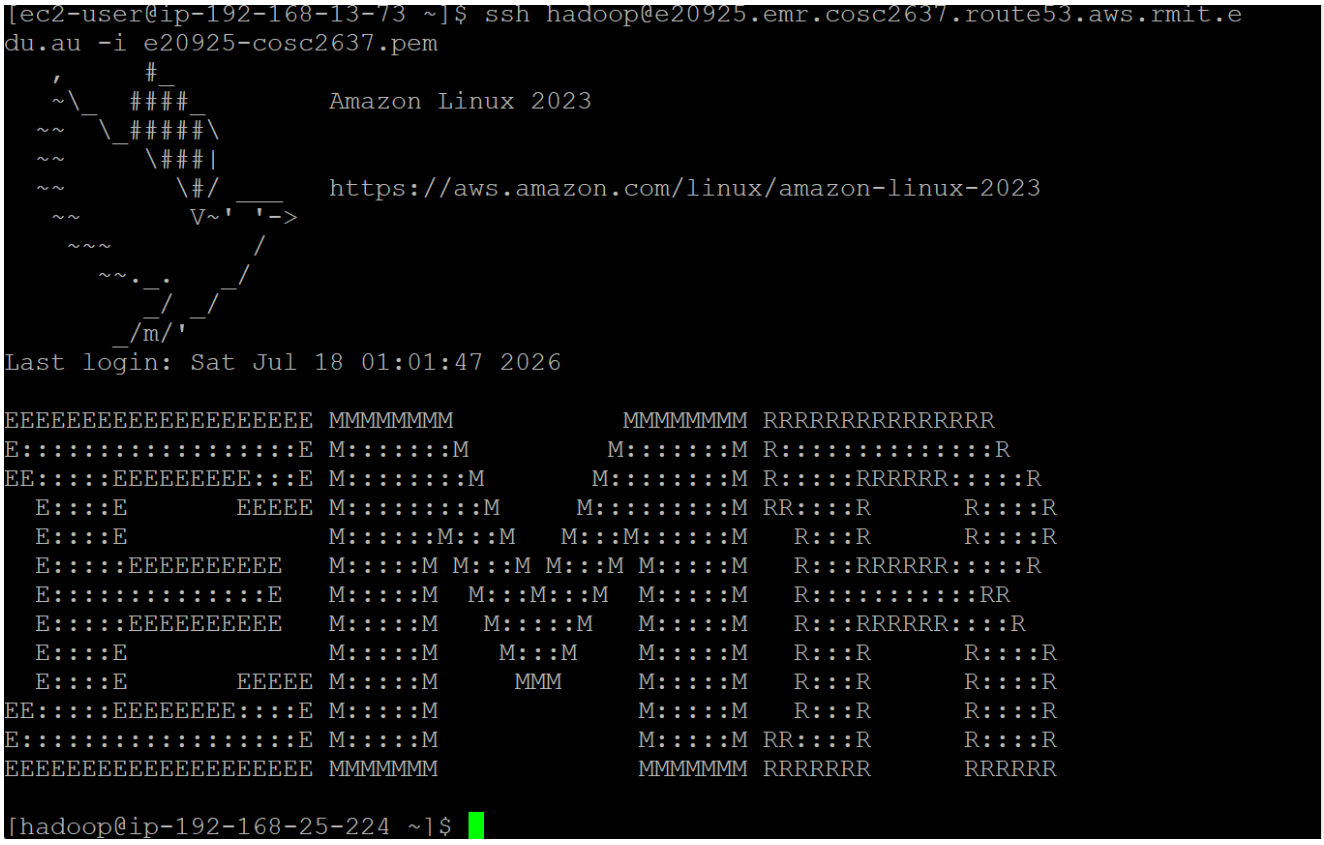
```
[ec2-user@ip-192-168-13-73 ~]$ ll
total 20
-rwxr-xr-x. 1 root     root     1674 Jul 16 03:05 create_cluster.sh
drwxrwxr-x. 2 ec2-user ec2-user   78 Jul 17 11:49 data
-r--------. 1 ec2-user ec2-user 1675 Apr 28  2023 e20925-cosc2637.pem
-rw-r--r--. 1 ec2-user ec2-user  132 Jul 17 10:04 instructions
-rw-r--r--. 1 root     root        7 Jul 17 06:36 student
-rwxr-xr-x. 1 root     root      651 Jun 19  2020 terminate_cluster.sh
[ec2-user@ip-192-168-13-73 ~]$
```

And login the newly created cluster by ssh hadoop@sxxxx.emr.... as shown below

```
[ec2-user@ip-192-168-13-73 ~]$ ./create_cluster.sh
Checking if EMR Cluster exists ...
EMR cluster already exists.

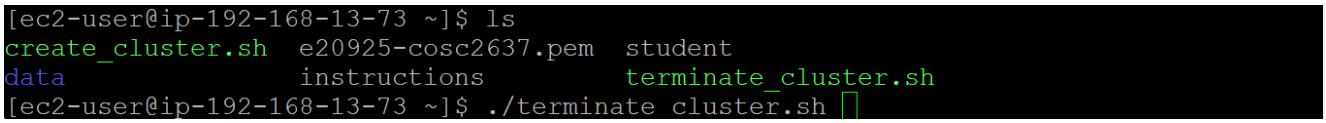
Access EMR Master node via SSH from your jumphost
ie ssh hadoop@e20925.emr.cosc2637.route53.aws.rmit.edu.au -i e20925-cosc2637.pem
[ec2-user@ip-192-168-13-73 ~]$ ls
create_cluster.sh  e20925-cosc2637.pem  student
data               instructions         terminate_cluster.sh
[ec2-user@ip-192-168-13-73 ~]$ ll
total 20
-rwxr-xr-x. 1 root     root     1674 Jul 16 03:05 create_cluster.sh
drwxrwxr-x. 2 ec2-user ec2-user   78 Jul 17 11:49 data
-r--------. 1 ec2-user ec2-user 1675 Apr 28  2023 e20925-cosc2637.pem
-rw-r--r--. 1 ec2-user ec2-user  132 Jul 17 10:04 instructions
-rw-r--r--. 1 root     root        7 Jul 17 06:36 student
-rwxr-xr-x. 1 root     root      651 Jun 19  2020 terminate_cluster.sh
[ec2-user@ip-192-168-13-73 ~]$ ssh hadoop@e20925.emr.cosc2637.route53.aws.rmit.e
du.au -i e20925-cosc2637.pem
```

If logging in successfully, you will see



Leave the cluster by inputting “exit”, then you will return back to Jumphost.

Don’t forget to shut down the cluster using terminate_cluster.sh. The shutting down will take about 5 minutes.

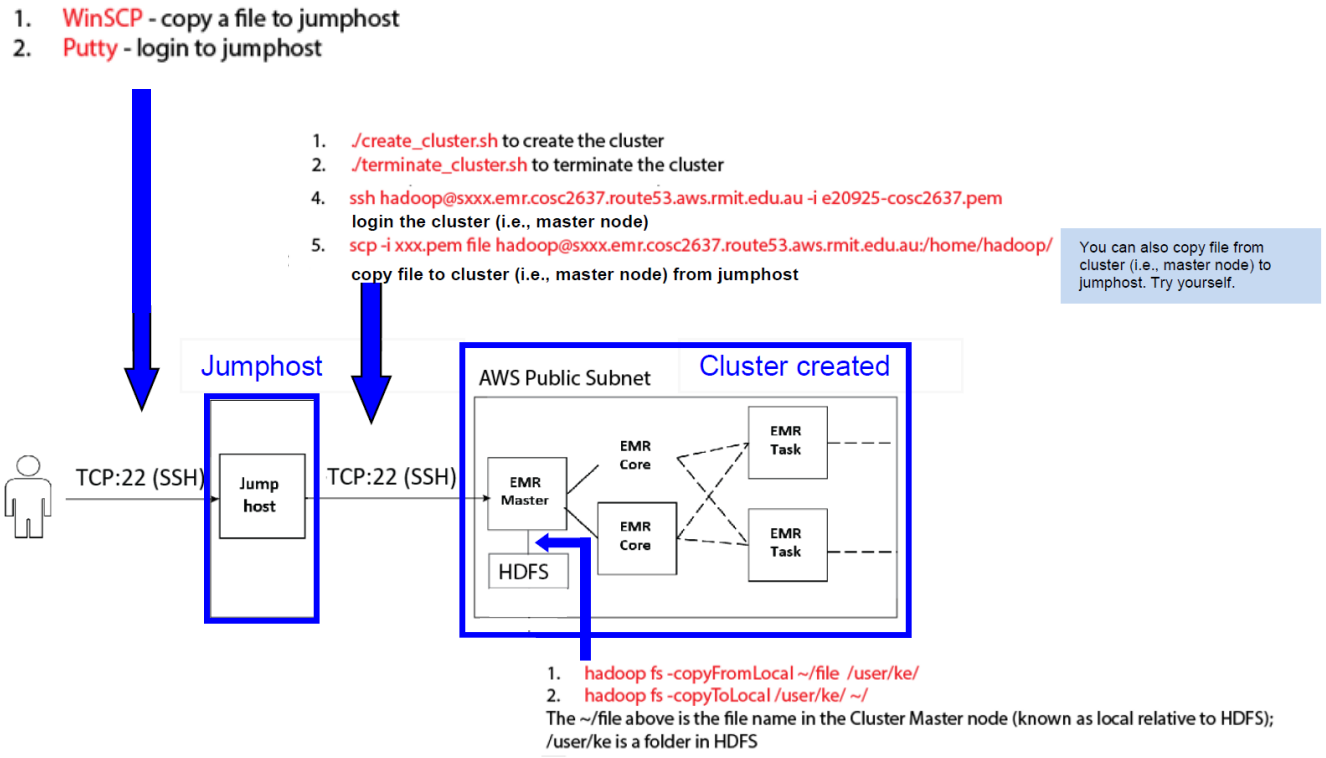


*********Useful Information**********

It is a good idea to do just prior to practical class so that it is up and running. However, do NOT leave it running for days at a time as it is expensive and a waste of resources to have idle clusters that are not being used. When finished, don't forget to **shutdown the cluster** to avoid charges. Note that there is no indication when the cluster is terminated.

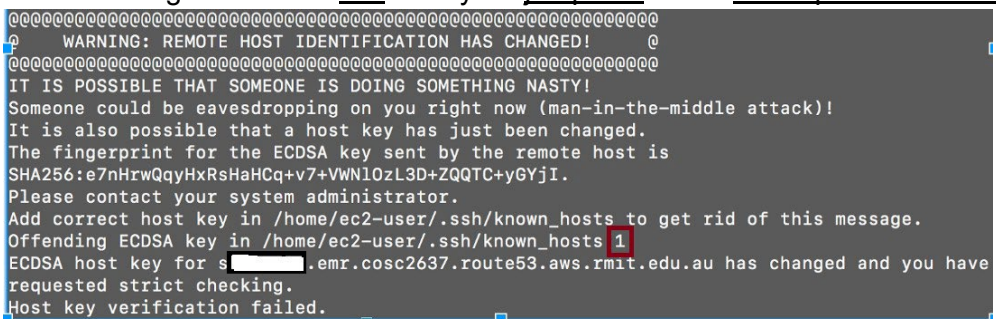
AWS EMR - Cheat sheet





A common issue of cluster login

You may see the following error when ssh from your jumphost to the hadoop master node,



Every time a new cluster is created it is a new set of hosts so the ssh host key changes, and ssh gives you a warning that it has been changed. To fix the problem, please run the following:

```
$ssh-keygen -R sxxxxxxx.emr.cosc2637.route53.aws.rmit.edu.au
```